Development

Requirements: Node.js 20+, Python 3.10+, and for the browser tests Playwright with Chromium.

Layout

```
SCW1/      the map: sources/, ontology.spec.json, maxey_registry.jsc
tools/     build_scw1.py (the map builder), new_packs_v4.py (generat
server/    the MCP server (Cloudflare Worker) and its tests
app/       build_ui.py, source/ (fixed input page), the built app, b
plugin/    the Claude plugin
docs/      documentation and images
```

Build

```
python tools/build_scw1.py build      # the map → SCW1/ and server/s
python app/build_ui.py                # the app → app/maxey0-context
```

Both are deterministic: the same inputs give byte-identical outputs.

Test

```
cd server
npm install
npm test                               # server and map tests

npx wrangler dev --port 8794           # in a second terminal: the s
cd ../app
node test_ui.cjs                       # the app in Chromium
```

```
node test_host.cjs                    # the app inside a simulated
node screenshots.cjs                  # refresh docs/images
```

Deploy

```
cd server
npx wrangler login                    # once
npx wrangler deploy
```

The Worker bundles the map and the app page at deploy time; rebuild both first when either changed. Check with `GET /` (version and digest) and `GET /app`.

Add or change a topic

1. Add or edit a pack in `SCW1/sources/packs/` (format: `human0.topic-pack/1`; the `topic-graph-generator` prompt describes it, and `validate_topic_pack` checks it).
2. Give the new topic a domain: add `"<topic id>": "<domain id>"` to `regroup.pack_domains` in `SCW1/ontology.spec.json`, or place its concepts in existing topics under `regroup.topics`. The build fails for any concept with no place.
3. Build, run the tests, deploy.

MCP server rows must be real projects whose existence you have checked; record where in the pack's `provenance.mcp_notes`.

Release

Raise the server version in `server/src/index.ts`, the plugin version in `plugin/maxey0-context-plane-graph/.claude-plugin/plugin.json`, and add an entry to `CHANGELOG.md`.